ESSENTIALS OF ARTIFICIAL INTELLIGENCE – CS22032



AI BASED INTELLIGENT MANGO QUALITY & RIPENESS ASSESSMENT SYSTEM
FINAL REPORT

Group Details		
Group Number	**Student Number**	**Student Name**
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1. ABSTRACT

Mango quality and ripeness assessment is an important task in the agricultural and fruit distribution industry because the quality and maturity of mangoes directly affect their market value, storage period, and suitability for consumption. Traditional assessment methods are mainly based on manual visual inspection, which can be subjective, time-consuming, and inconsistent between different individuals.

This project presents an AI-Based Intelligent Mango Quality & Ripeness Assessment System that uses computer vision and deep learning to automatically analyse mango images. The proposed system combines YOLOv8-based mango detection/segmentation, an EfficientNet-B0 convolutional neural network for classification, image-based colour analysis using HSV features, hybrid decision-making, and a rule-based recommendation system.

The system classifies input images into four categories: Grade A – Ripe/Fresh, Grade B – Unripe/Green, Grade C – Overripe/Damaged, and Non-Mango. Based on the predicted quality category, the system also provides practical information such as estimated price level, shelf-life guidance, storage recommendations, and handling recommendations.

The final model was evaluated using an untouched test set of 70 images. The system achieved an overall classification accuracy of 91.43%, with a mango-only accuracy of **91.18%** and a macro F1-score of 0.8591. The hybrid ripeness analysis achieved the same overall test accuracy as the CNN alone, indicating that the deep learning classifier was already effective while the additional image-analysis components primarily provide supporting information and decision validation.

The project demonstrates how AI-based image classification and intelligent decision-making can be applied to a practical agricultural problem. The system also highlights limitations related to image quality, difficult viewing angles, object detection, dataset size, and the reliability of automated decisions.

2. INTRODUCTION

Artificial Intelligence has become increasingly useful in solving real-world problems involving image recognition, classification, prediction, and decision-making. In agriculture and food distribution, computer vision can be used to analyse visual characteristics of agricultural products and support faster and more consistent quality assessment.

Mangoes undergo visible changes during ripening, including changes in colour, texture, surface condition, and the appearance of defects. These characteristics can potentially be analysed using computer vision and machine learning techniques.

The proposed project develops an intelligent system that accepts an image of a mango and automatically determines its quality and ripeness category. Instead of relying entirely on manual inspection, the system uses a combination of object detection, deep learning classification, image analysis, and rule-based decision-making.

The project follows the assignment requirement of developing a practical AI-based solution for a real-world problem and demonstrating how AI techniques can support classification and decision-making. The assignment guideline specifically requires students to analyse the problem, select and justify AI techniques, design the system, develop a prototype/model, and test the solution using sample inputs.

3. PROBLEM BACKGROUND

Current Problem

Mango quality assessment is commonly performed through manual visual inspection. A person may examine the colour, maturity, surface damage, spots, and general appearance of a mango before deciding whether it is suitable for sale or consumption.

However, manual inspection has several limitations:

- It can be time-consuming when many mangoes need to be assessed.
- Different people may make different judgments.
- Human assessment can be affected by lighting conditions and fatigue.
- Subtle differences between ripeness stages may be difficult to identify consistently.
- Manual grading is difficult to standardise across large quantities of fruit.
- Quality assessment can become inefficient in commercial environments.

Why AI Can Help
An AI-based image analysis system can process mango images automatically and identify visual patterns associated with different quality categories.

The proposed system can:

1. Detect the mango in an image.
2. Analyse its visual characteristics.
3. Classify its quality/ripeness category.
4. Reject images that are identified as non-mango objects.
5. Provide additional recommendations based on the classification.

This makes the system useful as a **decision-support tool**, rather than replacing human responsibility completely.

4. PROBLEM STATEMENT

Manual mango quality and ripeness assessment can be subjective, time-consuming, and inconsistent because it depends heavily on human visual inspection. There is a need for an intelligent computer vision-based system that can automatically analyse mango images and classify them according to ripeness and visible quality characteristics. The system should also provide useful recommendations based on the classification while maintaining reasonable accuracy and clearly communicating its limitations.

5. OBJECTIVES

Main Objective
To develop an AI-based intelligent system capable of automatically assessing mango quality and ripeness from images using computer vision and deep learning techniques.

Specific Objectives
1. To detect mangoes in input images using an object detection/segmentation model.
2. To classify mango images into Grade A – Ripe/Fresh, Grade B – Unripe/Green, Grade C – Overripe/Damaged, and Non-Mango.
3. To use deep learning to identify visual patterns associated with mango quality and ripeness.
4. To incorporate image-based colour and surface analysis to support the classification process.
5. To develop a hybrid decision-making process combining model predictions and image characteristics.

6. To provide practical recommendations such as price level, shelf-life guidance, storage conditions, and handling recommendations.
7. To evaluate the performance of the proposed system using a separate test dataset and real application inputs.
8. To identify limitations and potential risks associated with automated mango quality assessment.

6. SELECTED AI TECHNIQUES

YOLOv8 Object Detection/Segmentation
YOLOv8 is used to locate mangoes within an input image. The purpose is to identify the relevant mango region before classification, reducing the influence of unnecessary background information. The segmentation version of YOLOv8 additionally provides a mask representing the detected mango region.
System Workflow: Input image → YOLOv8 → detected mango region → classifier

Convolutional Neural Network – EfficientNet-B0
EfficientNet-B0 is the main deep learning classification model. It receives the cropped mango image and predicts one of the four classes:

- Grade A – Ripe/Fresh
- Grade B – Unripe/Green
- Grade C – Overripe/Damaged
- Non-Mango

EfficientNet is suitable because image classification requires learning complex visual patterns such as colour distribution, texture, shape, surface appearance, visual signs of ripeness, and damage/defects. The model uses transfer learning from an ImageNet-pretrained EfficientNet-B0 and adapts the final classification layer for the four project-specific classes.

Image-Based HSV Colour Analysis
The system also analyses the mango image using HSV colour information (Hue, Saturation, Value). This provides additional information about colour characteristics associated with mango ripeness (e.g., greener vs. yellower regions). However, HSV analysis is not treated as the main classifier because lighting conditions can affect colour measurements.

Hybrid Decision-Making
The system combines deep learning prediction with additional image-based evidence including CNN classification probabilities, HSV colour information, and surface defect information. The deep learning model maintains the strongest influence, while the hybrid

component acts primarily as a supporting and validation layer.

Rule-Based Decision System
After AI classification, the system applies predefined rules to convert the classification into practical recommendations:

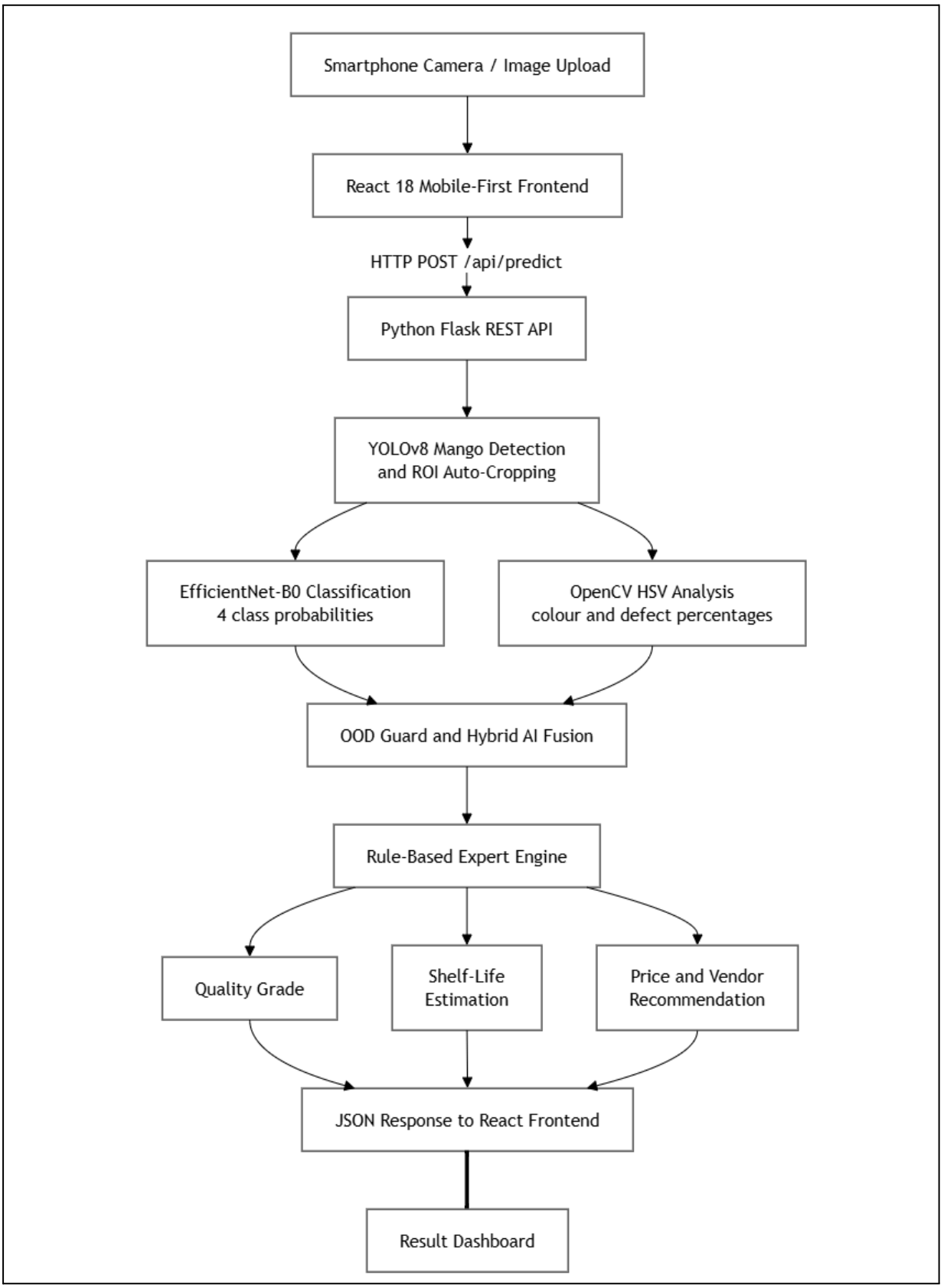
- **Grade A:** Price Level 100%, Shelf Life 3–5 days, Recommendation: Ready for sale/consumption
- **Grade B:** Price Level 90%, Shelf Life 7–10 days, Recommendation: Allow further ripening
- **Grade C:** Price Level 50%, Shelf Life ~1 day, Recommendation: Immediate clearance/processing
- **Non-Mango:** Price Level Invalid, Shelf Life N/A, Recommendation: Reject input

7. JUSTIFICATION OF AI TECHNIQUES

Technique	**Purpose**	**Why Selected**
YOLOv8 Segmentation	Detect and isolate mango	Automatically identifies the relevant object and reduces background interference
EfficientNet-B0 CNN	Classify mango quality/ripeness	Suitable for complex visual classification and transfer learning
HSV Analysis	Analyse colour characteristics	Provides interpretable colour-based evidence
Hybrid Decision-Making	Combine different evidence	Provides additional validation rather than relying on one signal
Rule-Based Reasoning	Generate recommendations	Converts classification results into practical decisions

8. SYSTEM DESIGN

System Architecture



9. SYSTEM WORKFLOW

1. **Image Input:** User uploads an image through the web interface.
2. **Mango Detection:** YOLOv8 identifies the mango and obtains the region/mask.
3. **Image Preprocessing:** Detected region is resized and normalized for classification.
4. **Deep Learning Classification:** EfficientNet-B0 processes the image and computes class probabilities.
5. **Computer Vision Analysis:** HSV colour features and surface defect metrics are extracted.
6. **Hybrid Analysis:** CNN prediction is evaluated alongside supporting visual evidence.
7. **Rule Engine:** Predefined rules convert the result into commercial and storage guidance.
8. **Final Output:** Web UI presents the classification grade, confidence score, and recommendations.

10. IMPLEMENTATION DETAILS

Technologies Used

- **Frontend:** React, HTML/CSS, JavaScript
- **Backend:** Python, Flask
- **AI/Computer Vision:** PyTorch, Torchvision, EfficientNet-B0, Ultralytics YOLOv8, OpenCV, NumPy
- **Development/Training:** Google Colab (GPU-accelerated), GitHub

11. DATASET

Dataset Classes

1. Grade A – Ripe/Fresh
2. Grade B – Unripe/Green
3. Grade C – Overripe/Damaged
4. Non-Mango

Dataset Split	Number of Images	Purpose
Training	546	Model Training
Validation	136	Model Validation and Tuning
Training + Validation	682	Model Development
Final Test Set	70	Final Performance Evaluation

Data Preparation

Images were prepared using preprocessing and augmentation pipelines including resizing, cropping, horizontal flipping, rotation, colour jitter, affine transformations, perspective transformations, and normalization to improve generalization.

12. MODEL TRAINING

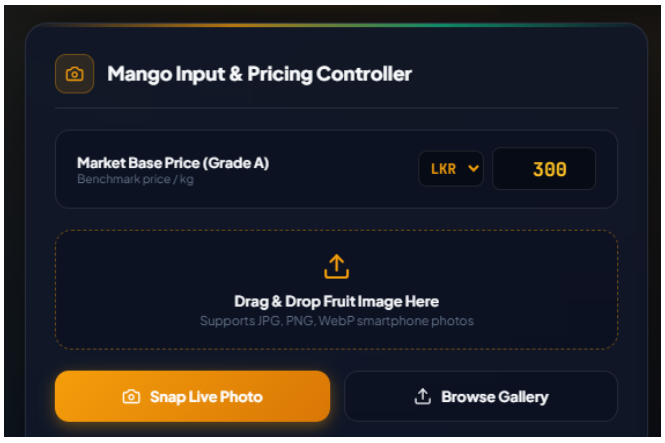
The classification model uses EfficientNet-B0 pre-trained on ImageNet. Training techniques included:

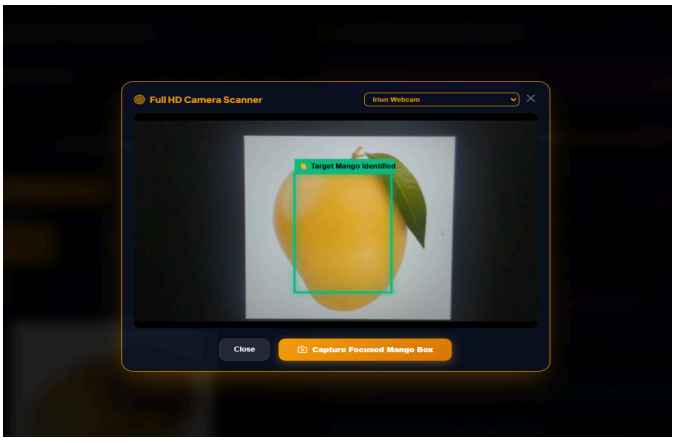
- Transfer learning
- Data augmentation
- Weighted sampling & Focal loss
- Staged training & Early stopping
- Learning-rate scheduling

13. USER INTERFACE

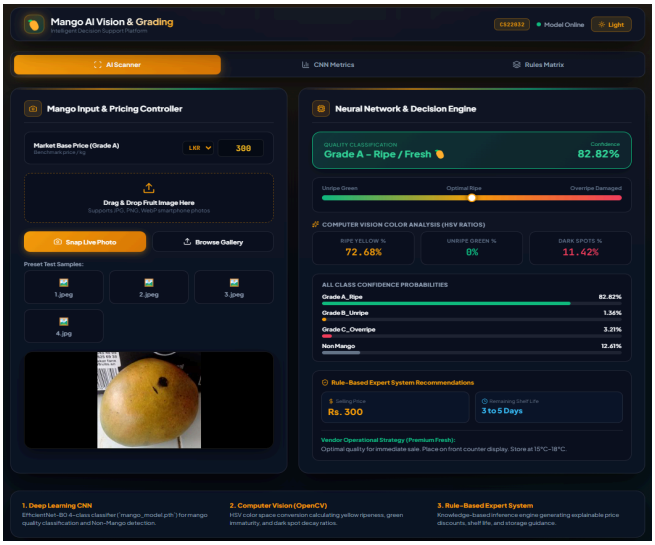
The application interface consists of:

- **Home / Upload Interface:** Allows users to upload or take a photo of a mango.

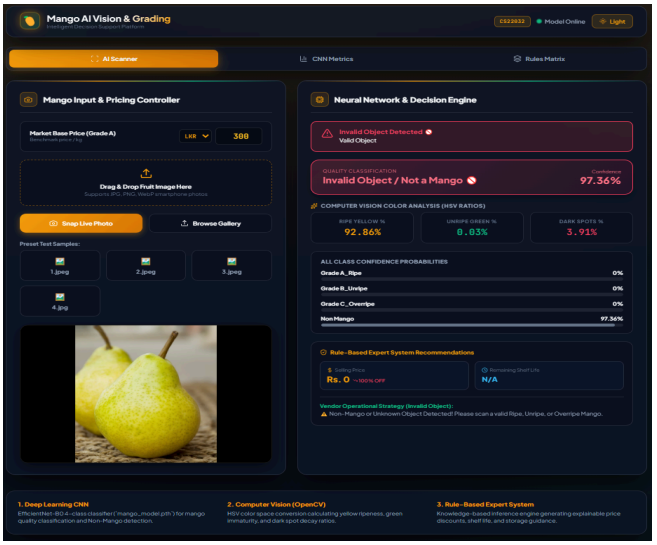




- **Result Dashboard:** Displays detected bounding box/mask, predicted grade (Grade A/B/C or Non-Mango), confidence score percentage, price level recommendation, shelf life estimation, and handling advice.



14.



TESTING AND RESULTS

Testing Strategy

The final application was evaluated using black-box testing directly against the finalized Flask API without modifying the AI models or training pipeline. Testing consisted of functional testing, AI model evaluation, end-to-end system testing, and edge-case/error analysis. No models were retrained and no AI logic was modified during testing.

Functional Testing

Functional verification was performed to validate backend API responsiveness, upload handling, classification routing, error handling, and output completeness.

Test ID	Test Case	Expected Result	Actual Result	Status
FT-01	API health check	Server online and API available	Status online, models loaded; HTTP 200 OK	PASS
FT-02	Valid mango image upload	Image accepted and processed	Grade A result returned; Success=True	PASS
FT-03	Clear Grade A / Ripe	Grade A classification	Grade_A_Ripe	PASS

Test ID	Test Case	Expected Result	Actual Result	Status
FT-04	Clear Grade B / Unripe	Grade B classification	Grade_B_Unripe	PASS
FT-05	Clear Grade C / Overripe	Grade C classification	Grade_C_Overripe	PASS
FT-06	Non-mango image	Non-Mango classification	Grade_B_Unripe	PASS
FT-07	Invalid text file format	Error response	Error handled correctly; 500 response	PASS
FT-08	Response field verification	Required fields present	All required fields found	PASS

Seven out of eight functional tests passed (87.5%). FT-06 was the only failed functional test because the tested Non-Mango image was incorrectly classified as Grade B – Unripe.

AI Model Evaluation

Evaluation was conducted on an untouched test set of 70 images.

- **Overall Accuracy:** 91.43%
- **Mango-only Accuracy:** 91.18%
- **Macro F1-score:** 0.8591

The confusion matrix summarizing class-level performance across the test set is presented below:

Classification Confusion Matrix

The complete production pipeline was evaluated across the full application path: Image Input → React Frontend → Flask API → YOLOv8 → EfficientNet-B0 → HSV/Defect Analysis → Hybrid Analysis → Rule Engine → Final Result.

Actual \ Predicted	Grade A	Grade B	Grade C	Non-Mango
Grade A	22	0	0	0
Grade B	1	23	0	1
Grade C	3	0	17	1

Actual \ Predicted	Grade A	Grade B	Grade C	Non-Mango
Non-Mango	0	0	0	2

End-to-End System Testing

Test ID	Condition	Expected Result	Actual Result	Status
E2E-01	Clear ripe mango	Grade A	Grade_A_Ripe, confidence 89.39%	PASS
E2E-02	Clear unripe mango	Grade B	Grade_B_Unripe, confidence 58.09%	PASS
E2E-03	Overripe/damaged mango	Grade C	Grade_C_Overripe, confidence 69.54%	PASS
E2E-04	Non-mango	Non-Mango	Non_Mango, confidence 100%	PASS
E2E-05	Poor lighting	Grade B	Grade_B_Unripe, confidence 86.12%	PASS
E2E-06	Blurry/low-quality	Grade A	Grade_A_Ripe, confidence 94.7%	PASS
E2E-07	Difficult background	Grade C	Grade_C_Overripe, confidence 72.58%	PASS
E2E-08	Difficult viewing angle	Grade A	Grade_A_Ripe, confidence 90.56%	PASS

Hybrid Model Evaluation

To evaluate the impact of visual computer vision features on deep learning prediction accuracy, performance was compared between the standalone CNN and the full hybrid system on the 70-image test set.

Model	Test Accuracy
Pure CNN (EfficientNet-B0)	91.43%

Model	Test Accuracy
Hybrid System (CNN + HSV + Defects)	91.43%

The hybrid approach achieved the same overall accuracy as the pure CNN on the untouched test set. Therefore, the hybrid component did not provide a measurable improvement in raw classification accuracy. It was retained as a supporting analysis and validation layer.

- CNN correct → Hybrid wrong: 0
- CNN wrong → Hybrid correct: 0

Error and Edge-Case Analysis

Misclassifications within the 70-image set primarily occurred between Grade C (Overripe) and Grade A (Ripe) due to subtle visual overlaps under ambient lighting conditions. In addition, a specific edge-case test was conducted outside the evaluation dataset:

- **Image:** dataset/val/Non_Mango/non_mango_wiki_0.jpg
- **Actual Class:** Non-Mango
- **Predicted:** Grade_B_Unripe
- **Error Type:** False positive
- **Likely Reason:** The object visually resembled a green unripe mango in colour and shape, causing the computer-vision detection and EfficientNet classification to produce an incorrect mango prediction.

Testing Summary

Testing Category	Result
Functional Testing	7/8 passed (87.5%)
AI Model Accuracy	91.43%
Mango-only Accuracy	91.18%
Macro F1-score	0.8591
Hybrid Accuracy	91.43%
End-to-End Testing	8/8 passed
Edge-Case Testing	1 authentic false-positive identified

Overall, the testing demonstrates that the proposed system performs effectively for the tested

mango quality categories and application scenarios. The AI model achieved 91.43% accuracy on the untouched test set, while all eight selected end-to-end cases produced expected results. However, functional and edge-case testing identified a false-positive Non-Mango classification, demonstrating that objects with mango-like visual characteristics can still challenge the system.

15. LIMITATIONS AND DISCUSSION

1. **Dataset Size:** Test evaluation used 70 images; expansion to broader cultivars is needed.
2. **Image Quality & Lighting:** Blur, low resolution, or harsh shadows affect HSV metrics.
3. **Viewing Angle:** Partial visibility or extreme angles can hinder object detection.
4. **YOLO Detection Sensitivity:** Classification depends directly on successful mango localization.
5. **Rule-Based Assumptions:** Recommendation figures (pricing, shelf-life) are heuristic project rules.

16. ETHICAL AND SOCIAL CONSIDERATIONS

- **Accuracy & Reliability:** System predictions should serve as decision support rather than authoritative grading.
- **Human Responsibility:** Quality assurance and commercial transactions require human oversight.
- **Dataset Bias:** Model performance depends on representative training across various mango species.
- **Transparency:** Results explicitly display confidence scores to signal uncertainty.

17. CONCLUSION

The project successfully implemented an **AI-Based Intelligent Mango Quality & Ripeness Assessment System** combining YOLOv8, EfficientNet-B0, HSV feature extraction, and rule-based decision support. Achieving an overall accuracy of 91.43% on a 70-image test set, the system demonstrates the practical viability of computer vision for agricultural quality grading while identifying clear pathways for future enhancement.

18. REFERENCES

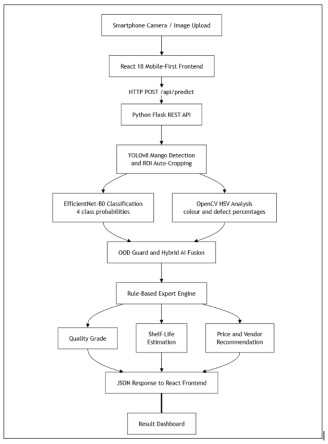
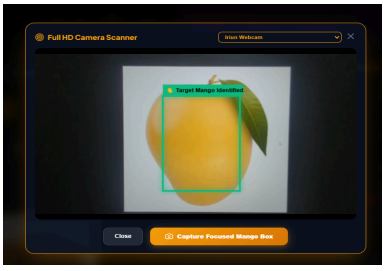
1. Tan, M., & Le, Q. V. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. *International Conference on Machine Learning (ICML)*.
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3. Bradski, G. (2000). The OpenCV Library. *Dr. Dobb's Journal of Software Tools*.
4. Paszke, A., et al. (2019). PyTorch: An imperative style, high-performance deep learning library. *Advances in Neural Information Processing Systems (NeurIPS)*.

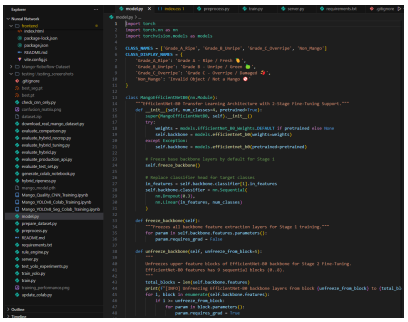
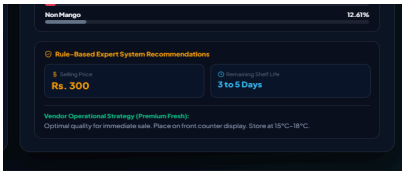
19. INDIVIDUAL CONTRIBUTION

INDIVIDUAL CONTRIBUTION SHEET- MEMBER 01

Student Name	Kavindu Kavishka	**Registration No.**	D/COE/25/0004
Degree / Intake	COE - Intake 42	**Group No.**	39
Project Title	AI Based Intelligence Mango Quality & Ripeness Assessment System		

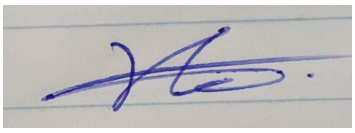
Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
System Architecture and Integration	Designed the overall system architecture and integrated the main components.	System Design, Component Integration	
Mango Detection and Segmentation	Integrated mango detection and segmentation into the system.	YOLOv8	

Quality Classification	Implemented the classification pipeline for four mango quality categories.	EfficientNet-B0, Deep Learning	
Colour and Defect Analysis	Developed colour analysis, defect analysis, and hybrid AI decision-making components.	HSV Colour Analysis, Computer Vision, Hybrid AI	
Recommendation and Price Estimation	Implemented rule-based logic for price estimation, remaining shelf life, and handling recommendations.	Rule-Based Decision Logic	
Backend and Frontend Integration	Developed the Flask REST API and integrated the backend with the React frontend.	Flask, REST API, React	
Testing and Validation	Participated in model training, debugging, performance evaluation, and final technical validation.	Model Training, Performance Evaluation, Debugging	

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

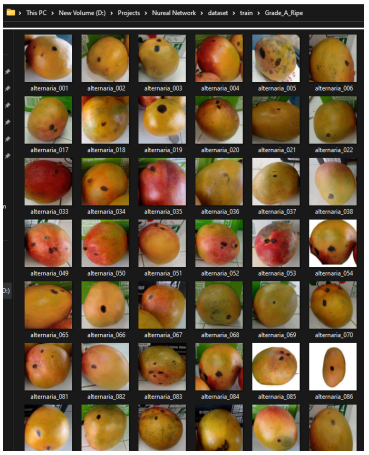
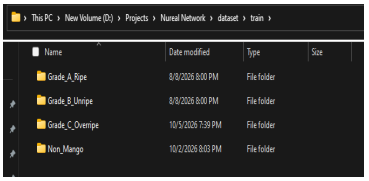
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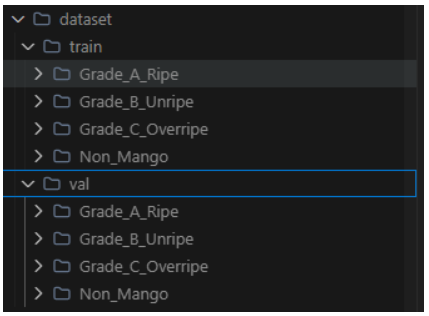
Date : 17/10/2026

INDIVIDUAL CONTRIBUTION SHEET- MEMBER 02

Student Name	O Fernando	**Registration No.**	D/DBA/25/0011
Degree / Intake	DBA - Intake 42	**Group No.**	39
Project Title	AI Based Intelligence Mango Quality & Ripeness Assessment System		

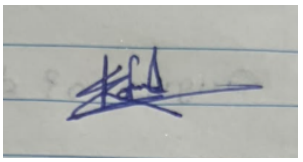
Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
Dataset Preparation	Contributed to the preparation and organization of the mango image dataset.	Image Dataset Preparation, Data Organization	
Quality Category Review	Reviewed the mango quality and ripeness categories used by the system.	Category Definition, Classification Requirements	
Image Quality Checking	Assisted in checking image quality and consistency across classification categories.	Image Inspection, Data Quality Checking	

Data Processing Support	Supported dataset preparation and data processing activities for model development.	Data Processing, Image Data Handling	
Computer Vision Feature Review	Reviewed computer vision feature requirements, including colour-based analysis.	Computer Vision, HSV Colour Analysis	
Dataset Organization and Validation	Contributed to ensuring the dataset organization supported the intended classification workflow.	Dataset Organization, Classification Workflow Review	

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

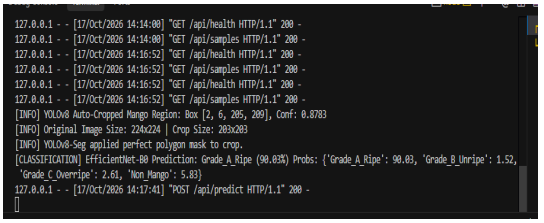
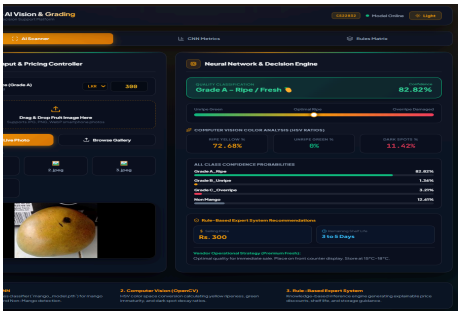
Student Signature: 

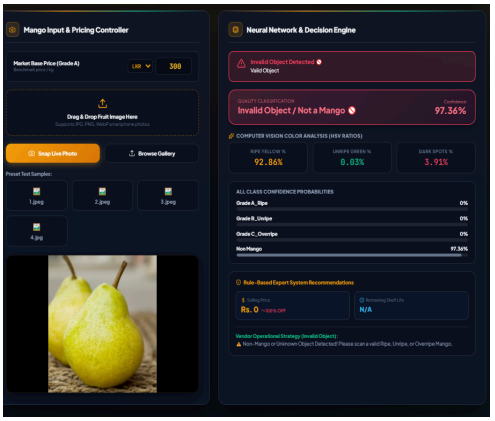
Date : 17/10/2026

INDIVIDUAL CONTRIBUTION SHEET- MEMBER 03

Student Name	HU Wijewickrama	**Registration No.**	D/BIT/24/0034
Degree / Intake	BIT - Intake 41	**Group No.**	39
Project Title	AI Based Intelligence Mango Quality & Ripeness Assessment System		

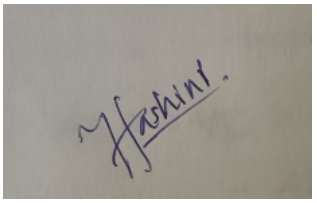
Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
Backend Testing	Contributed to backend and system-level technical validation.	Backend Testing, System Validation	
API Endpoint Testing	Tested API endpoints using sample inputs.	Flask REST API, Sample Input Testing	
Response and Error Handling	Checked JSON responses and API error-handling behaviour.	JSON, API Error Handling	
Prediction Verification	Verified prediction results and rule-based recommendation outputs.	AI Model Predictions, Rule-Based Logic	

Edge Case Analysis	Assisted in identifying and analysing system edge cases and failure conditions.	Edge Case Testing, Failure Analysis	
End-to-End System Validation	Contributed to checking the reliability of the complete prediction workflow.	End-to-End Testing, System Validation	

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

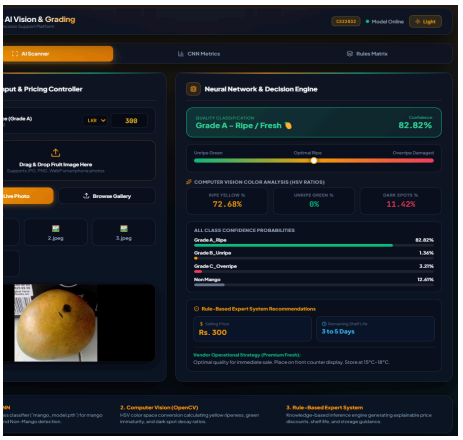
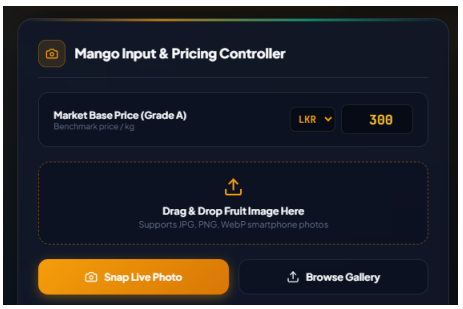
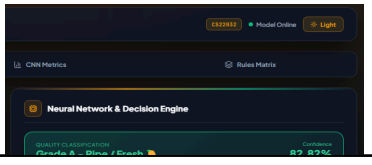
Student Signature: 

Date : 17/10/2026

INDIVIDUAL CONTRIBUTION SHEET- MEMBER 04

Student Name	ARK Vidurusinghe	**Registration No.**	D/BIT/24/0037
Degree / Intake	BIT - Intake 41	**Group No.**	39
Project Title	AI Based Intelligence Mango Quality & Ripeness Assessment System		

Technical Contributions:

Task / Component	My Contribution	Tools / AI Technique Used	Evidence / Output
User Interface Evaluation	Evaluated the React-based user interface and its main features.	React, UI Evaluation	
Image Upload and Camera Testing	Tested image-upload and camera-interaction functionality.	Image Upload, Camera Integration Testing	
Result Dashboard Evaluation	Reviewed the result dashboard and	React, Dashboard Evaluation	

	presentation of AI prediction results.		
Workflow Usability Testing	Evaluated the usability of the image submission and prediction workflow.	Usability Testing, Workflow Evaluation	
User Interface Improvement	Provided technical feedback on possible user interface improvements.	UI Review, Functional Testing	
Results and Recommendations Presentation	Evaluated how clearly classification results and recommendations were displayed to users.	UI Evaluation, Information Presentation	

Declaration

I confirm that the contributions listed above accurately represent my individual work in this group project, and I can explain or demonstrate the stated work if required.

Student Signature: 

Date : 17/10/2026